Hydro Power

- **Hydropower** is energy that comes from the force of moving water.
- Hydropower is called a **renewable** energy source because the water on Earth is continuously replenished by precipitation.
- As long as the water cycle continues, we won’t run out of this energy source.

- The total installed power generation capacity in India as on 31st January 2025 is 466,250.79 MW
- https://en.wikipedia.org/wiki/List_of_power_stations_in_India#Hydroelectric
- India's economically exploitable and viable hydroelectric potential is estimated to be 148,701 MW.
- An additional 6,780 MW from smaller hydro schemes (with capacities of less than 25 MW) is estimated as exploitable.
- 56 sites for pumped storage schemes with an aggregate installed capacity of 94,000 MW have also been identified.
- In central India, the hydroelectric power potential from the Godavari, Mahanadi, Nagavali, Vamsadhara and Narmada river basins has not been developed on a major scale due to potential opposition from the tribal population.

- **Hydro Dams**
- **Hydropower Plants**

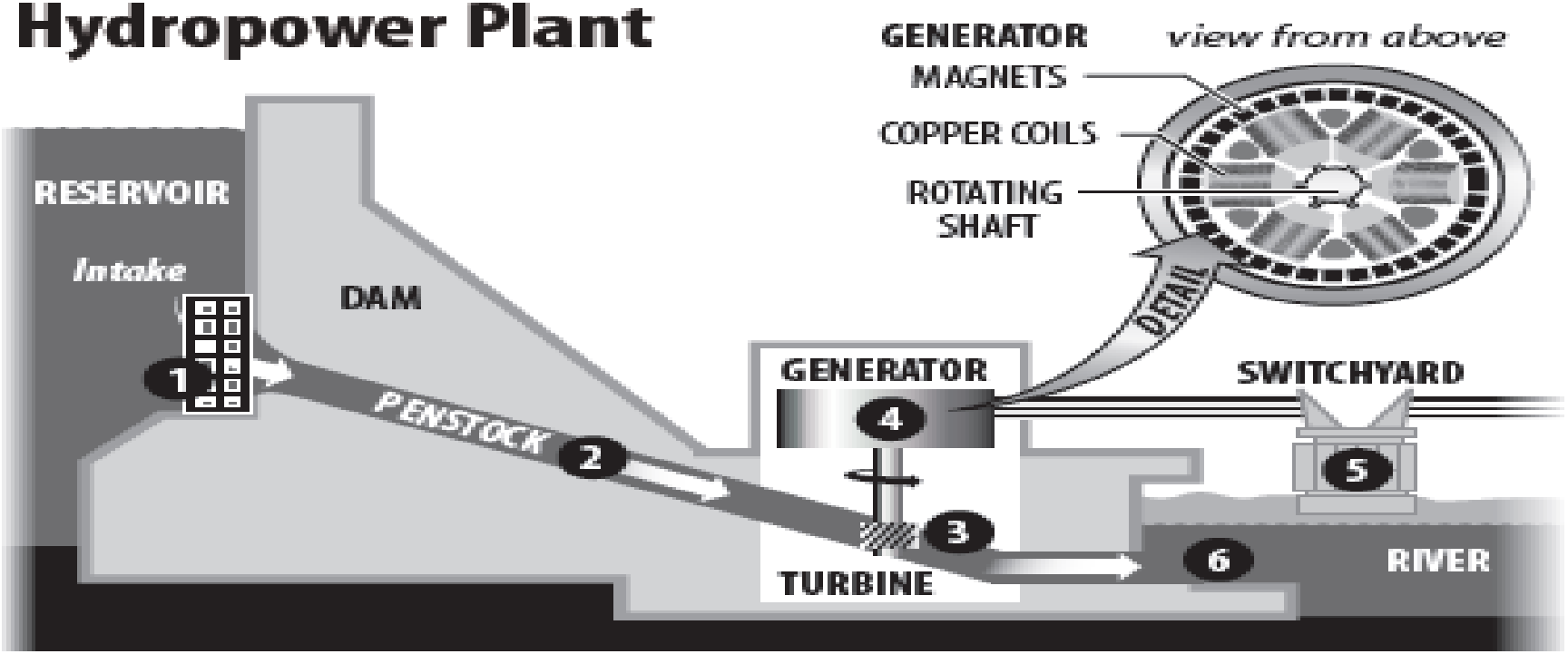
Hydro Dams

- A hydroelectric dam is a large structure that uses the energy of falling water to generate electricity.
- Dams are built across rivers to create reservoirs that store water.
- How it works
- Water is released from the reservoir and flows through a turbine, which spins a generator to produce electricity.
- The amount of electricity generated depends on the height of the dam and how fast water flows through the turbines.
- The electricity is then fed into the electrical grid to power homes and businesses.
- Uses of dams
- Dams can also be used for flood control, irrigation, water supply, and recreation.

Hydropower Plants

- A hydropower plant uses the energy of moving water to generate electricity. It's a renewable energy source that's been used for a long time.
- How it works
- A dam or diversion structure alters the flow of a river
- Water is stored in a reservoir
- Water is released from the reservoir and flows through a turbine
- The turbine spins and activates a generator to produce electricity
- The electricity is fed into the electrical grid

Hydropower Plant



1. Water in a reservoir behind a hydropower dam flows through an intake screen, which filters out large debris, but allows fish to pass through.
2. The water travels through a large pipe, called a penstock.
3. The force of the water spins a turbine at a low speed, allowing fish to pass through unharmed.
4. Inside the generator, the shaft spins coils of copper wire inside a ring of magnets. This creates an electric field, producing electricity.
5. Electricity is sent to a switchyard, where a transformer increases the voltage, allowing it to travel through the electric grid.
6. Water flows out of the penstock into the downstream river.

Advantages

- No fuel required
- No air pollution
- Can easily work during high peak daily loads
- Prevents floods

Disadvantages

- Disrupts the aquatic ecosystems
- Disruption of surrounding areas
- Requires large areas
- Large scale human displacement

Hydel scheme

1 **.Run-off Plants without Poundage**:

As name indicates this type of plant doesn't store water, the plant uses as water comes.

2. **Run-0ff plants with Poundage**:

Poundage permits storage of water during the off –peak period and use of this water during peak periods.

3. **Reservoir Plants**: A reservoir plant is that which has reservoir of such size as to permit carrying over storage from wet season to the next dry season.

4 .**Low head plants:** In this case small dam is built across the river to provide the necessary head. In such plants Francis type of turbines are used.

5. **Medium head plants**: The fore bay provided at the beginning of Penstock serves as water reservoir for such plants. In these plants water is generally carried out in open canals from reservoir to the Fore bay and then to the penstock.

6. **High head Plant**: This plants works above 500mtrs and Pelton wheel turbines are commonly used. In this plant water is carried out from the main reservoir by a tunnel up to surge tank and then from the surge tank to the power house in penstock.

7. **Base Load Plants**: These Plants are mainly depending on the nature of load. Is demand is more, this plants are used regularly and load factor of this plants are high.

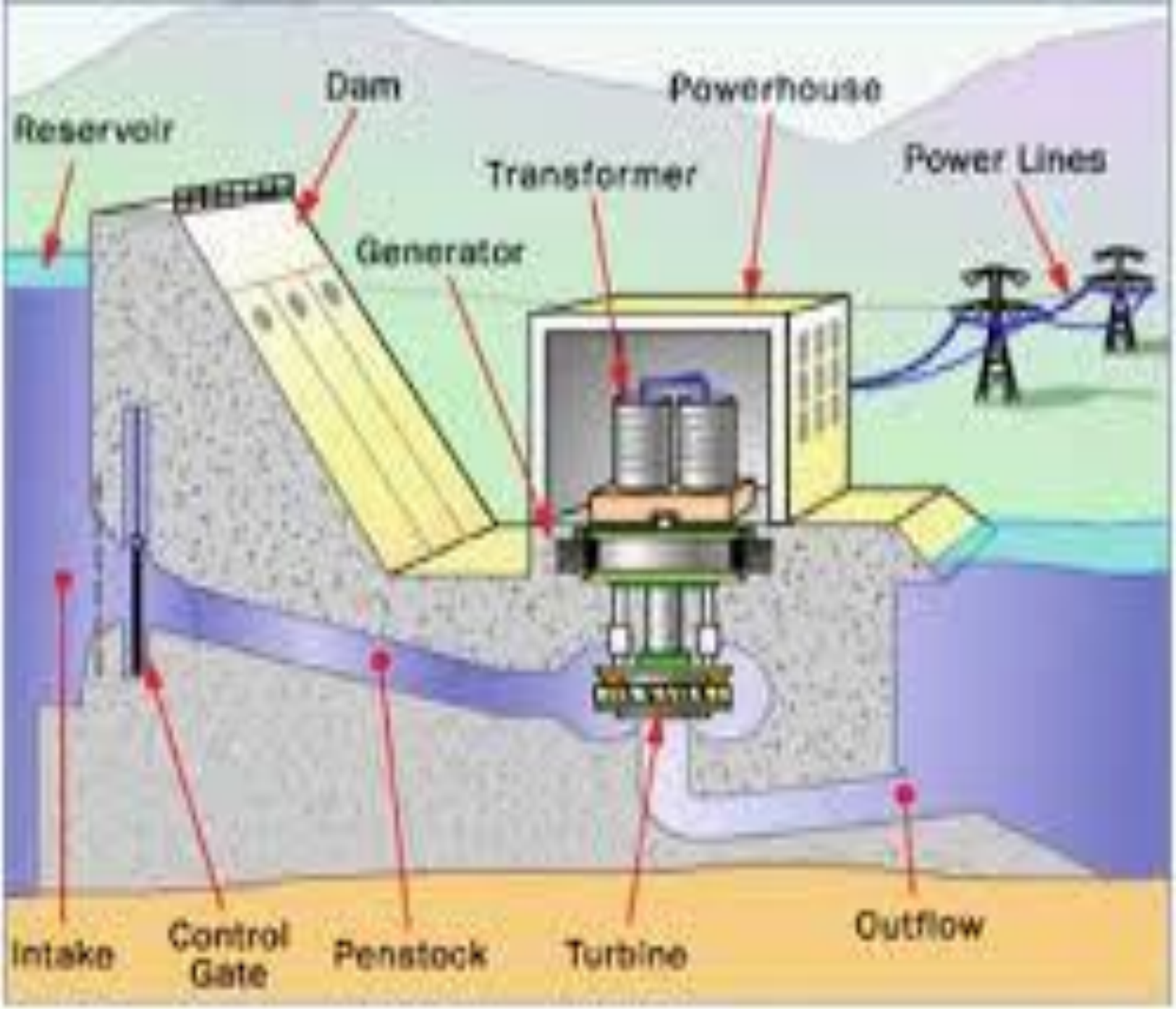
8. **Peak load Plants**: These plants are mainly used during the peak load. Run-off river plants with poundage can be used as peak-load plants. reservoir plants with enough storage behind the dam can be used either as base load or as peak load plants as required.

9. **Pumped storage plants**: These plants are used when quantity of water available for generation is insufficient. If it is possible to pond at head water and tail water locations after passing through the turbine is stored in the tail race pond from where it may be pumped back to the Head water pond.

Runoff River Type Hydropower Plant

Chief Joseph Dam near Bridgeport, Washington, USA.. Major 'Run-of-the-river' station without a sizeable reservoir

https://upload.wikimedia.org/wikipedia/commons/6/6c/Chief_Joseph_Dam.jpg


Reservoir
Dam
Powerhouse
Transformer
Power Lines
Generator
Intake
Control Gate
Penstock
Turbine
Outflow

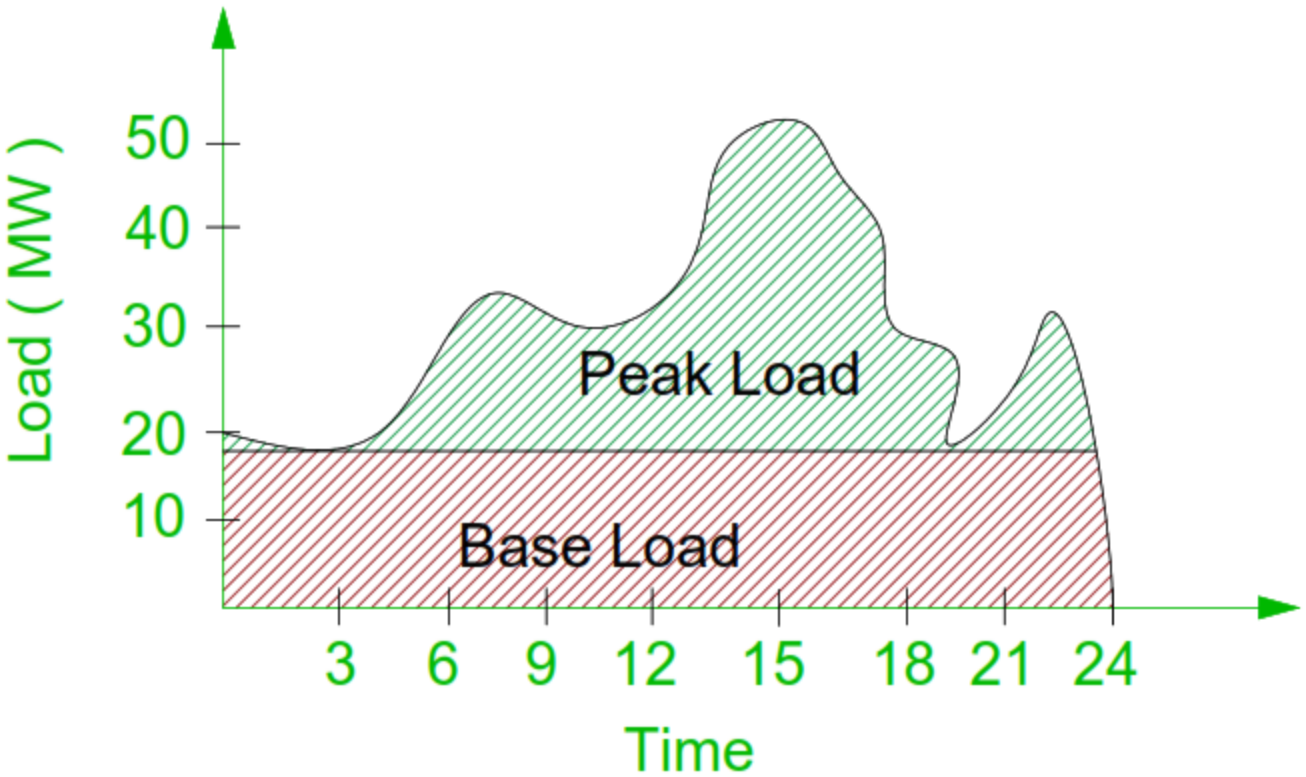


FIG : LOAD CURVE

PUMPED HYDROPOWER STORAGE GUIDE

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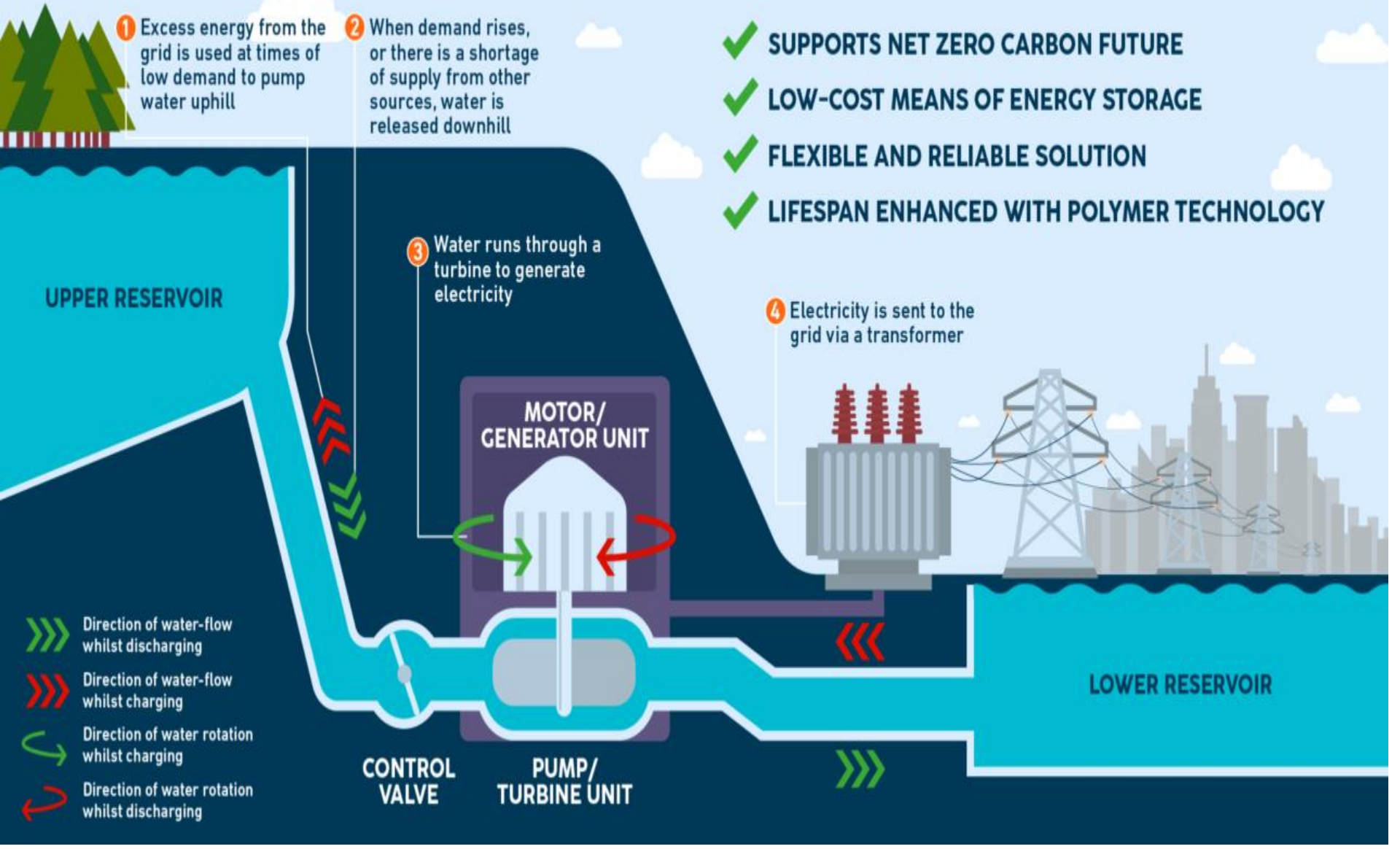
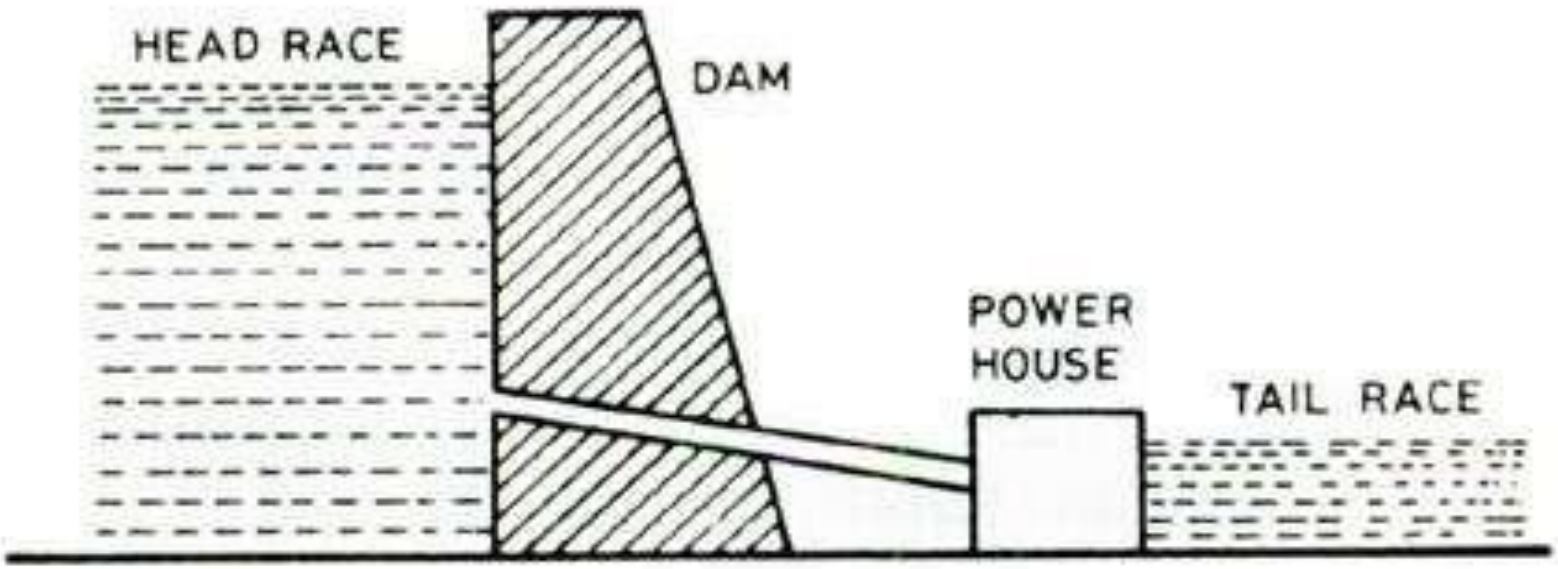


Low head plants

- In this case a small dam is built across the river to provide the necessary head.
- The excess water is allowed to flow over the dam itself.
- In such plants Francis, Propeller or Kaplan types of turbines are used.
- Also no surge tank is required.
- These plants are constructed where the water head available less then 30mtrs.
- The production of electricity will be less due to low head.

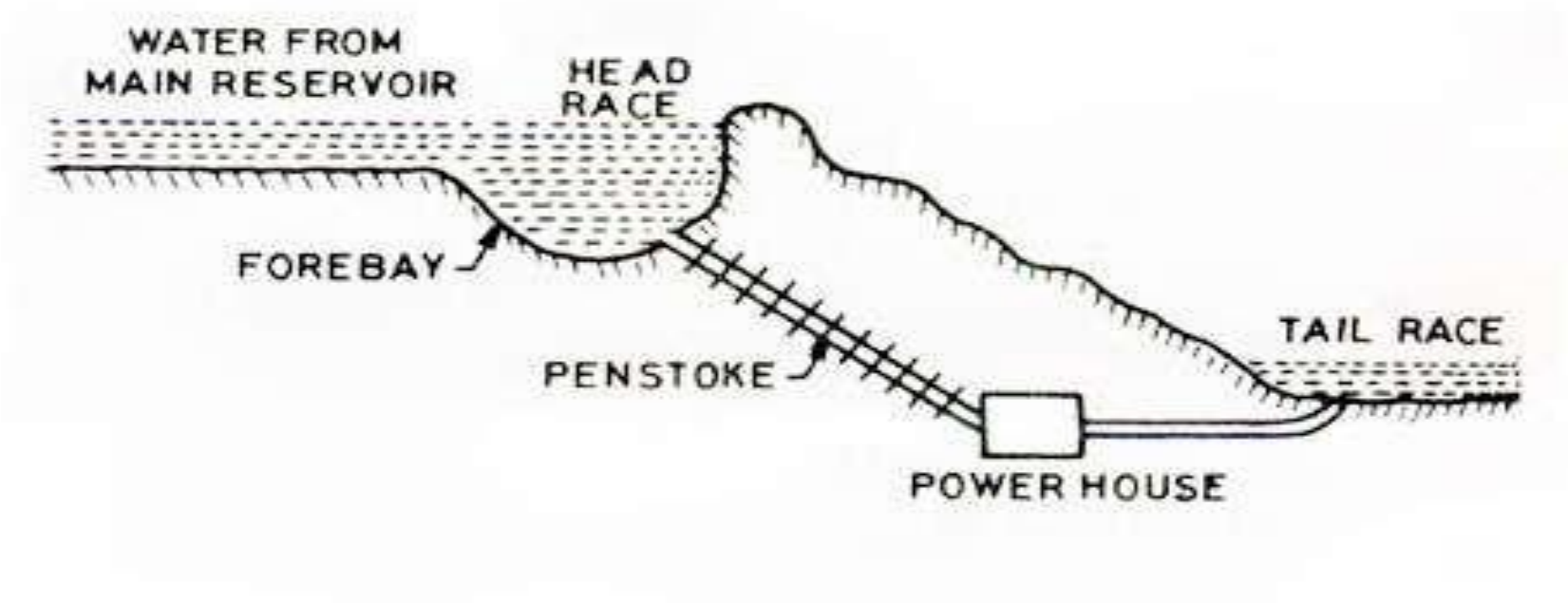
Low Head Plant



Medium head plants

- Mainly forebay provided before the Penstock, acts as water reservoir for medium head plants.
- In this plants mainly water is carried through main reservoir to forebay and then to the penstock.
- The forebay acts as surge tank for these plants.
- The turbines used will be Francis type of the steel encased variety.

Medium Head Plant



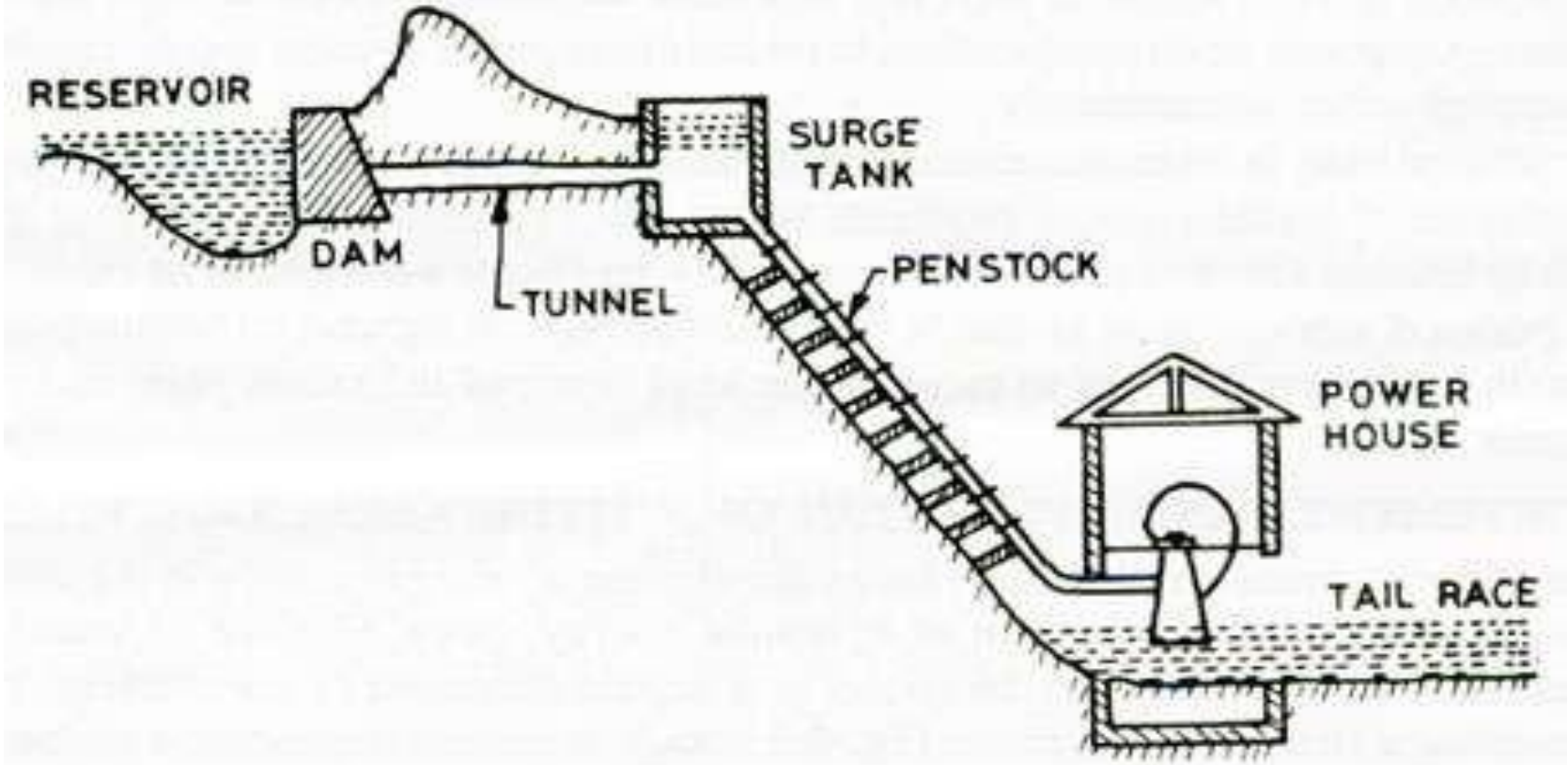
High head plants

- Mainly in these plants pressure tunnel is provided before the surge tank, which inturn connected to penstock.
- A pressure tunnel is taken off from the reservoir and water brought to the valve house at the start of the penstocks.
- The penstocks are huge steel pipes which take large quantity of water from the valve house to the power house.

- The valve house contains main sluice gates and in addition automatic isolating valves which come into operation when the penstock bursts, cutting further supply of water.
- Surge tank is an open tank and is built just in between the beginning of the penstocks and the valve house.
- In absence of surge tank, the water hammer can damage the fixed gates.

- In Majority of dams Sluice gates are provided.
- The sluice gates are opened when dam level is below level and there is shortage water for irrigation.
- Normally the high head plants are 500 meters above and for heads above 500 meters Pelton wheels are used.

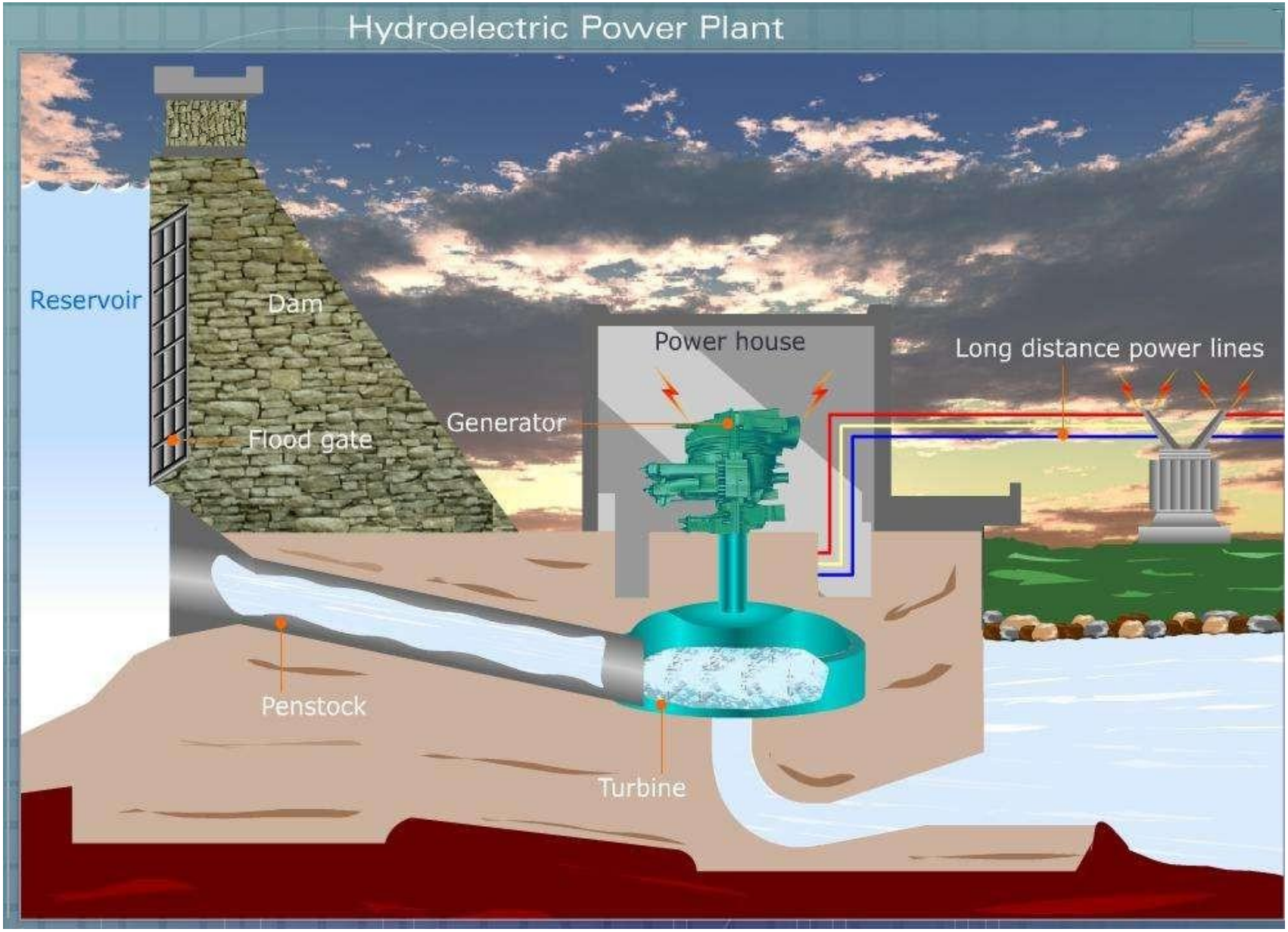
High Head Plant



Components of hydel scheme

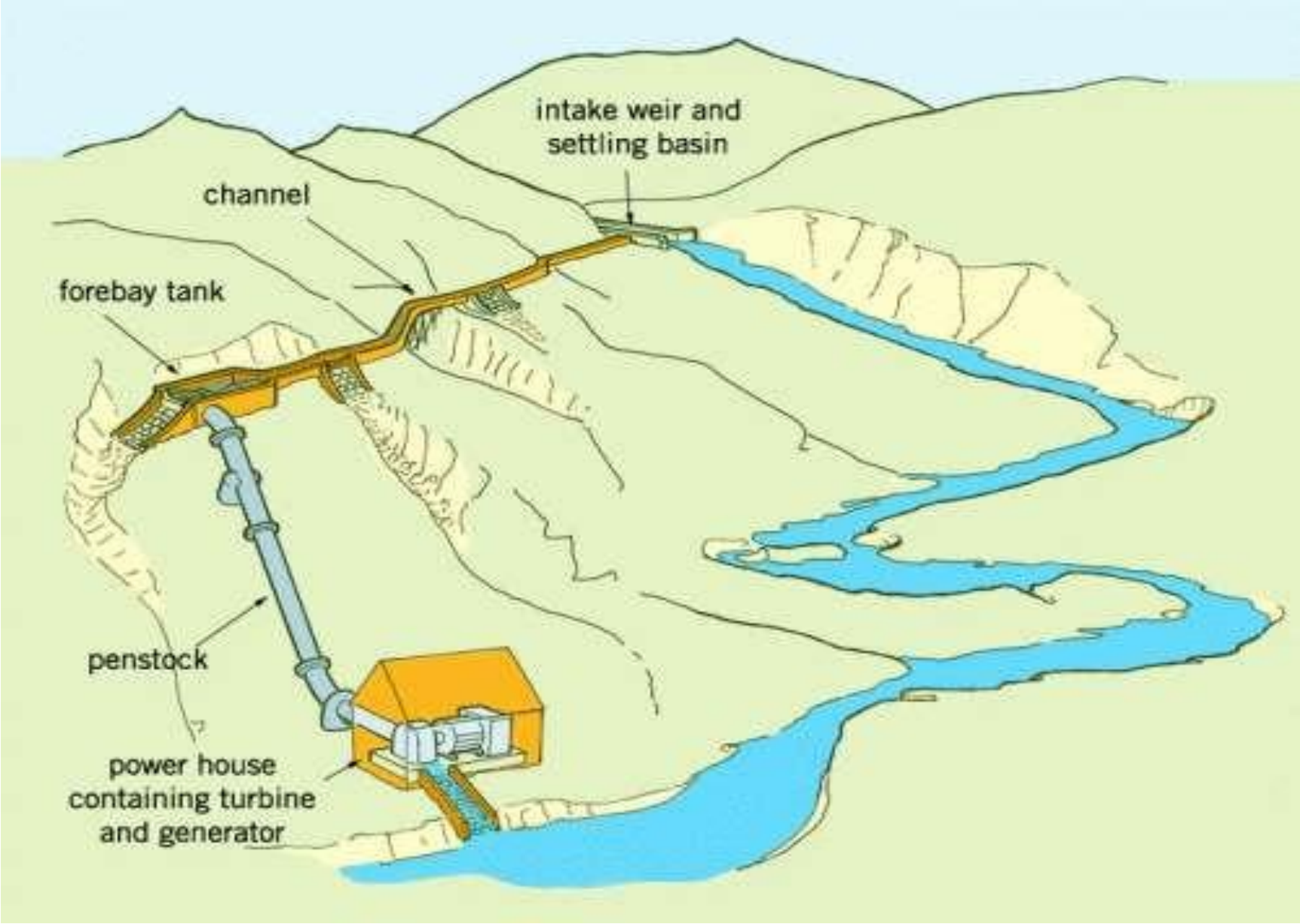
The principal components are:

1. Forebay
2. Intake structure
3. Penstocks
4. Surge tank
5. Turbines
6. Power house
7. Draft tube
8. Tail race


Hydroelectric Power Plant
Reservoir
Dam
Flood gate
Generator
Power house
Long distance power lines
Penstock
Turbine

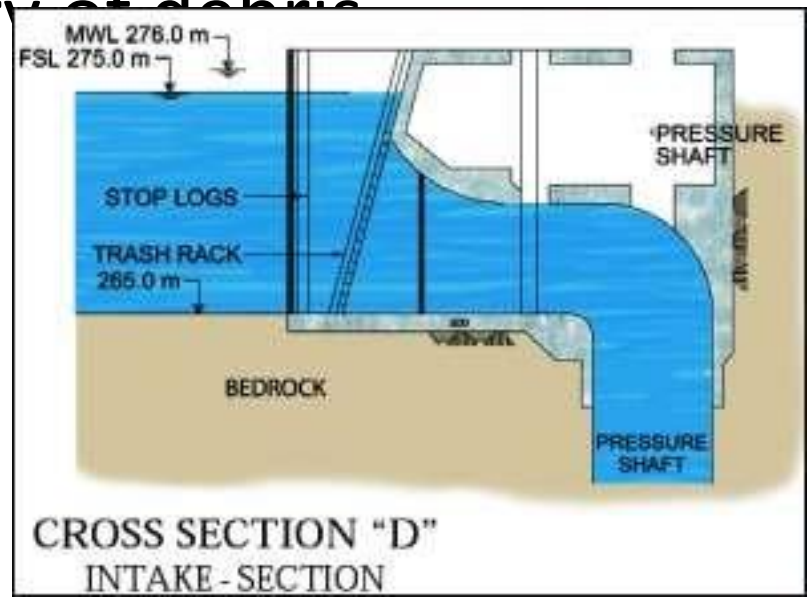
Forebay

- Enlarged body of water provided in front of penstock.
- Provided in case of run off river plants and storage plants.
- Main function to store water which is rejected by plant.
- Power house located closed to dam penstock directly take water from reservoir, reservoir act as forebay.


intake weir and
settling basin
channel
forebay tank
penstock
power house
containing turbine
and generator

Intake structure

- Water conveyed from forebay to penstocks through intake structures.
- Main components are trash rack and gate.
- Trash rack prevent entry of debris



Penstocks

- open or closed conduits which carry water to the turbines.
- made of reinforced concrete or steel. Concrete penstocks are suitable for low heads less then 30mtrs.
- steel penstocks are designed for any head.
- thickness of penstocks increases with head or water pressure

- penstocks gates are fixed to initial of penstocks, and flow of water is controlled by operating penstock gates.

- Either buried in ground or kept exposed.

Surge tank

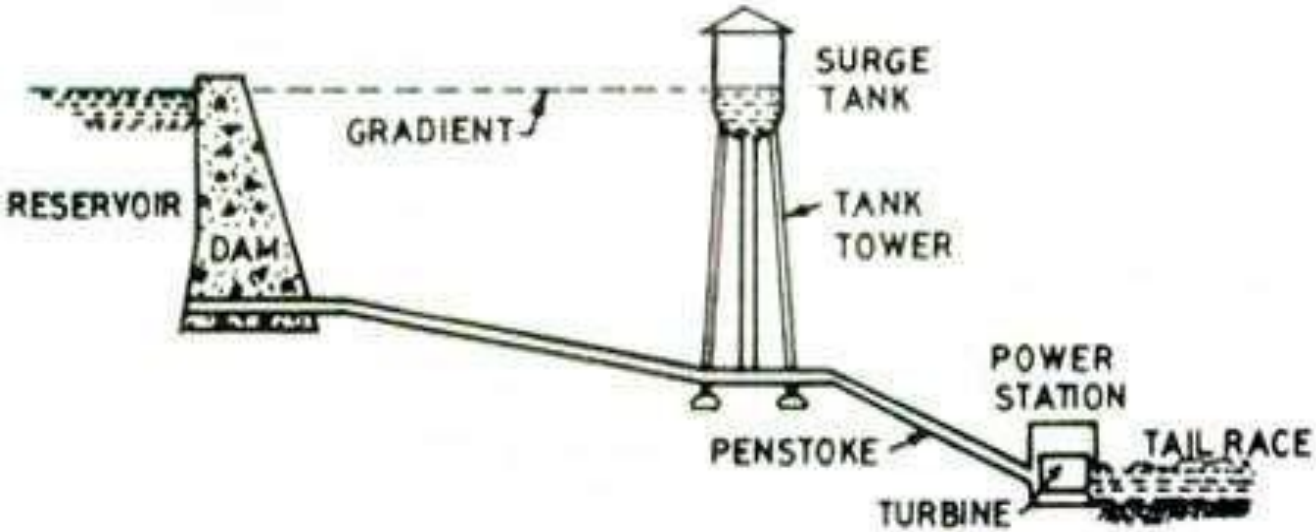
- additional storage for near to turbine, usually provided in high head plants.
- located near the beginning of the penstock.
- As the load on the turbine decreases or during load rejection by the turbine the surge tank provides space for holding water.

- surge tank over comes the abnormal pressure

  in the conduit when load on the turbine falls and acts as a reservoir during increase of load

  on t

Surge Tank



Turbines

- turbines are used to convert the energy water of falling water into mechanical energy.
- water turbine is a rotary engine that takes energy from moving water.
- flowing water is directed on to the blades of a turbine runner, creating a force on the

- Since the runner is spinning, the force acts through a distance n this way, energy is transferred from the water flow to the turbine.
- The principal types of turbines are:

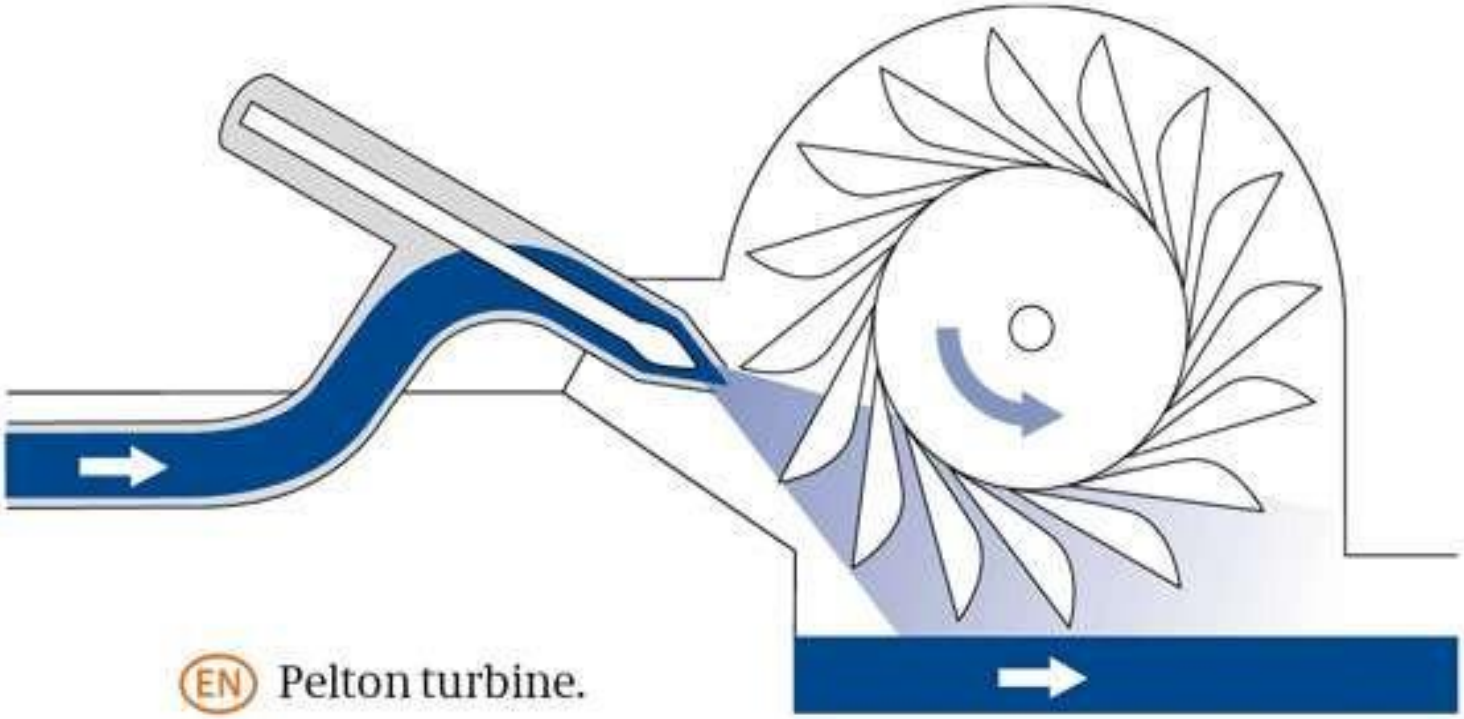
1) Impulse turbine
2) Reaction Turbine

Impulse turbines:

mainly used in high head plants.

- the entire pressure of water is converted into kinetic energy in a nozzle and the velocity of the jet drives the blades of turbine.
- The nozzle consist of a needle, and quantity of water jet falling on the turbine is controlled this needle placed in the tip of the nozzle.
- If the load on the turbine decreases, the governor pushes the needle into the nozzle, thereby reducing the quantity of water striking the turbine.

- Examples of Impulse turbines are:
- Pelton Wheel.
- Turgo
- Michell-Banki (also known as the Cross flow or Ossberger turbine.



EN Pelton turbine.

FR Turbine Pelton.

ES Turbina Pelton.

www.solarpraxis.de / M.Römer

Reaction turbines :

are mainly for low and medium head plants.

- In reaction turbine the water enters the runner partly with pressure energy and partly with velocity head.
- Most water turbines in use are reaction turbines and are used in low (<30m/98 ft) and medium (30-300m/98–984 ft)head applications.
- In reaction turbine pressure drop occurs in both fixed and moving blades.

- In this turbine the runner blades changed with respect to guide vane opening.
- As the sudden decrease of load takes place, the guide vane limit decreases according to that runner blade closes.

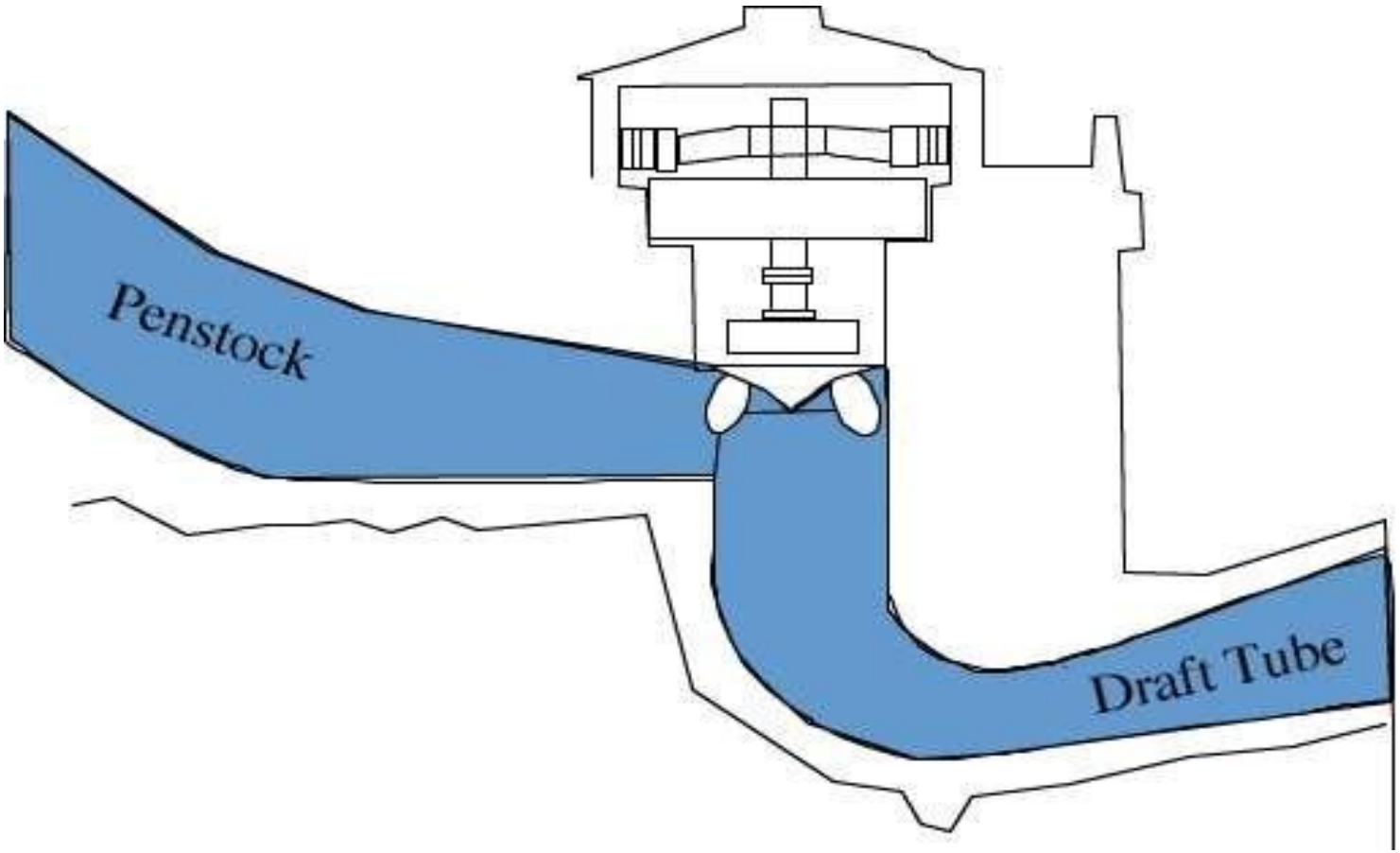
- Examples of reaction turbines are:
  - Francis turbine
  - Kaplan turbine

Francis

Kaplan

Draft tube

- is a pipe or passage of gradually increasing cross sectional area, which connect to the exit to tail race.
- it reduces high velocity of water discharged by the turbine.
- draft tube permits turbines to be installed at a higher level than the tail race level, which help the maintaince and repair of turbines.

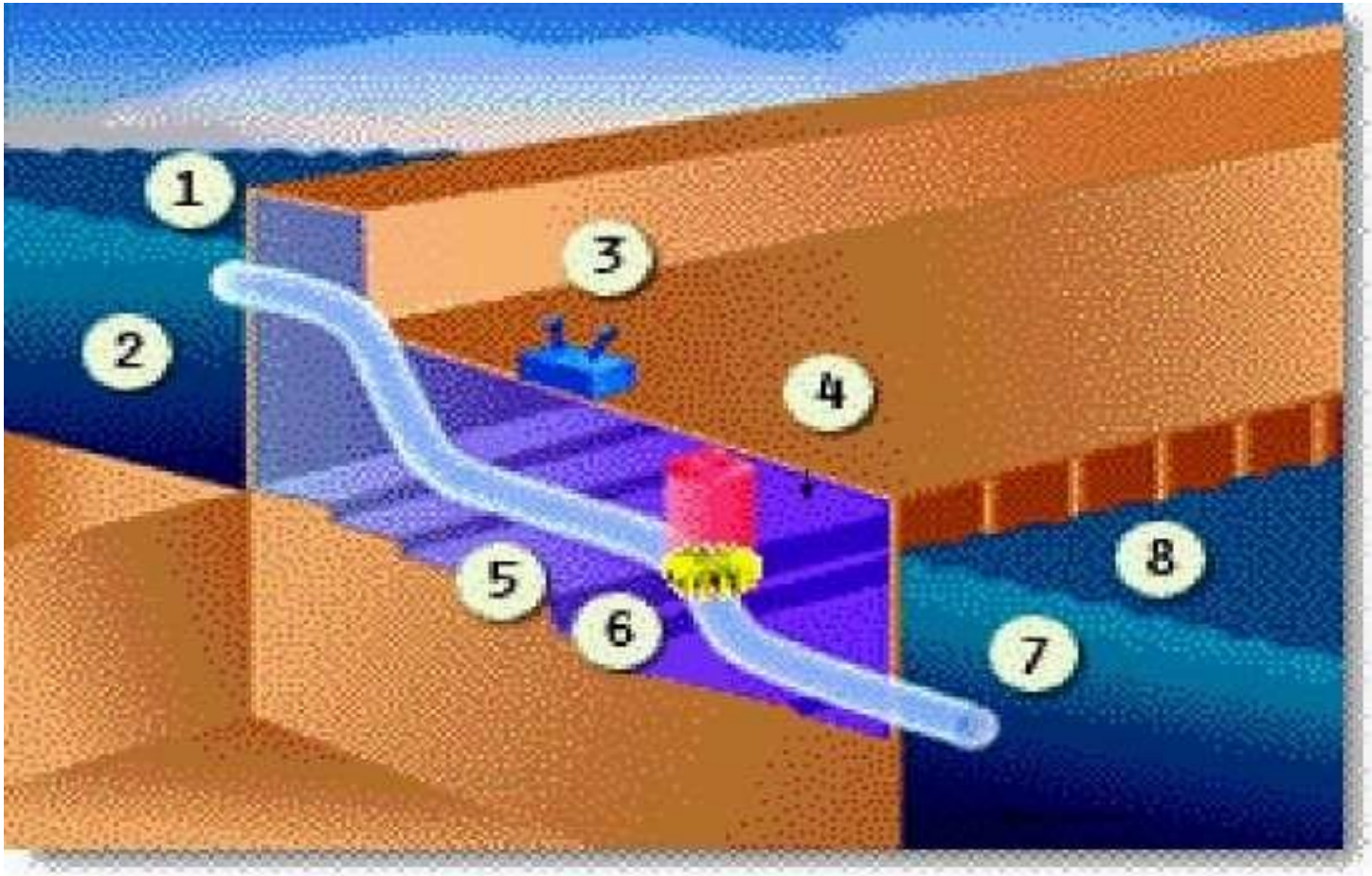

Penstock
Draft Tube

Power house

- Power house contains the electro mechanical equipment i.e. hydro power turbine, Generator, excitation system, main inlet valves, transformers, Switchyard, DC systems, governor, bus duct, step up transformers, step down transformers, high voltages switch gears, control metering for protection of systems.

Tail race

- tail race tunnel or channel are provided to direct the used water coming out of draft tube back to the river.
- important criteria of designing the tail race is kind of draft tube, the gross head and geographical situation of the area.
- Tail race is designed in such a way that water hammer is minimizes when water leaves the draft tube.


1
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8

Power generation

The amount of electricity that can be generated by a hydropower plant depends on two factors:

- **flow rate** - the quantity of water flowing in a given time; and
- **head** - the height from which the water falls.

The greater the flow and head, the more electricity produced.

Flow Rate = the quantity of water flowing
Head = the height from which water falls

A standard equation for calculating energy production:

$$\frac{\text{Power} = (\text{Head}) \times (\text{Flow}) \times (\text{Efficiency})}{11.8}$$

Power = the electric power in kilowatts or kW

Head = the distance the water falls (measured in feet)

Flow = the amount of water flowing (measured in cubic feet per second or **cfs**)

Efficiency = How well the turbine and generator convert the power of falling water into electric power. This can range from 60% (0.60) for older, poorly maintained hydroplants to 90% (0.90) for newer, well maintained plants.

11.8 = Index that converts units of feet and seconds into kilowatts

- A key new technology for hydro energy gaining traction in India is the "Surface Hydrokinetic Turbine (SHKT)," which harnesses the kinetic energy of flowing water in existing water infrastructure like canals and tailrace channels without the need for large dams, offering a cost-effective way to generate renewable energy with minimal environmental impact; the Central Electricity Authority (CEA) has recently recognized this technology as a potential solution for India's energy needs.

- Key points about SHKT technology:

- **No dam required:**
- Unlike traditional hydropower, SHKT generates electricity directly from the kinetic energy of flowing water, eliminating the need for large dams and associated environmental concerns.
- **Cost-effective:**
- SHKT is considered to be a cost-effective option, with electricity generation costs estimated at ₹2-3 per unit.
- **Scalable:**
- This technology can be scaled to generate significant amounts of power by utilizing existing water infrastructure like canals and tailrace channels.
- **Grid stability support:**
- SHKT can contribute to grid stability by providing base-load, round-the-clock renewable energy, particularly in areas with limited grid access.